

Classification of Elements and Periodicity in Properties

Complete chapter notes — the genesis of the table, its structure from electronic configuration, every periodic trend with its reason, and the anomalies that examiners live on.

The spine of this chapter: two quantities explain almost every trend — the **effective nuclear charge** Z_{eff} pulling the outer electrons in, and the **number of shells** pushing them out. Across a period Z_{eff} rises with no new shell; down a group a new shell is added and shielding grows. Every trend in this chapter is one of those two effects winning. Anomalies happen only where a **stable half-filled or fully-filled subshell**, or an unusually **small size**, interferes.

1 · Genesis of Periodic Classification

Attempt	Idea	Why it failed
Döbereiner's triads (1829)	In a group of three similar elements, the atomic mass of the middle one is roughly the mean of the other two. Li (6.9), Na (23.0), K (39.1): mean = 23.0 ✓	Only a handful of triads could be found
Newlands' law of octaves (1865)	Arranged by increasing atomic mass, every eighth element repeats the properties of the first, like the notes of a musical scale	Valid only up to calcium; no room for the elements discovered later
Mendeleev's periodic law (1869)	"The properties of elements are a periodic function of their atomic masses ." Arranged in 8 groups and 6 periods, giving priority to chemical similarity over mass order	See the drawbacks below

1.1 Mendeleev — merits and drawbacks

Merits. (i) He left **gaps** for undiscovered elements and predicted their properties with remarkable accuracy — *eka*-aluminium (gallium), *eka*-boron (scandium) and *eka*-silicon (germanium) were all found later and matched. (ii) He **corrected wrong atomic masses** (beryllium, indium, gold) on the strength of chemical behaviour. (iii) Noble gases, discovered much later, slotted in as a new group without disturbing the table.

DRAWBACKS YOU MUST BE ABLE TO LIST

1. **Position of hydrogen** was uncertain — it resembles both the alkali metals ($1s^1$, forms H^+) and the halogens (one electron short of a full shell, forms H^- , diatomic).
2. **Anomalous pairs.** To keep chemically similar elements together, he had to place a heavier element before a lighter one: Ar (39.9) before K (39.1), Co (58.9) before Ni (58.7), Te (127.6) before I (126.9).
3. **No place for isotopes** — they have different masses but must occupy the same position.
4. Chemically dissimilar elements were grouped together (alkali metals with coinage metals Cu, Ag, Au).

1.2 Moseley and the modern periodic law

Moseley (1913) found that the frequency of the characteristic X-rays emitted by an element obeys $\sqrt{\nu} = a(Z - b)$ — a clean relationship with the **atomic number**, not the atomic mass. Atomic number is the fundamental property, and the anomalous pairs disappear the moment the table is ordered by Z .

Modern periodic law: the physical and chemical properties of the elements are a **periodic function of their atomic numbers**.

2 · Structure of the Modern Periodic Table

The long form has **18 vertical groups** and **7 horizontal periods**. Periodicity arises for one reason only: the **recurrence of similar outer-shell electronic configurations** after regular intervals.

	group →																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1																		
2																		
3																		
4																		
5																		
6			*															
7			**															

*																		
**																		

s-block: ns^{1-2} · p-block: ns^2np^{1-6} · d-block: $(n-1)d^{1-10}ns^{0-2}$ · f-block: $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$

* lanthanoids (58–71, after La) · ** actinoids (90–103, after Ac) — taken out of period 6 and 7 to keep the t

Block structure of the periodic table. The block is named after the subshell that receives the last electron. Period length is fixed by which subshells are being filled, which is why the periods run 2, 8, 8, 18, 18, 32, 32.

Period	Subshells filled	Number of elements
1	1s	2
2	2s 2p	8
3	3s 3p	8
4	4s 3d 4p	18
5	5s 4d 5p	18
6	6s 4f 5d 6p	32
7	7s 5f 6d 7p	32

2.1 The four blocks

Block	Groups	General outer configuration	Characteristics
s	1, 2 (and He)	ns^{1-2}	Soft, highly reactive metals; low IE; strong reducing agents; lose electrons to form +1/+2 ions
p	13–18	ns^2np^{1-6}	Metals, non-metals and metalloids together; group 18 are the noble gases with filled shells
d (transition)	3–12	$(n-1)d^{1-10}ns^{0-2}$	Hard, high-melting metals; variable oxidation states; coloured ions; catalytic and magnetic behaviour
f (inner transition)	Placed separately	$(n-2)f^{1-14}(n-1)d^{0-1}ns^2$	Lanthanoids and actinoids; all metals; actinoids are radioactive

FREQUENTLY ASKED

Zinc, cadmium and mercury (group 12) have a completely filled d subshell in both the atom and all their common ions (d^{10}), so they are **not regarded as typical transition elements**. Helium is placed in group 18 despite the $1s^2$ configuration because its *properties* are those of a noble gas.

2.2 Locating an element from its configuration

Period number = the highest principal quantum number n in the configuration. **Group number:** s-block = number of valence electrons; p-block = $10 +$ number of electrons in $ns + np$; d-block = number of electrons in $(n-1)d + ns$.

WORKED EXAMPLE 1 — POSITION FROM CONFIGURATION

(a) $Z = 17$: $1s^2 2s^2 2p^6 3s^2 3p^5$. Highest $n = 3 \Rightarrow$ period 3; p-block with $2 + 5 = 7$ outer electrons $\Rightarrow$ group $10 + 7 =$ **17, period 3 (chlorine)**.

(b) $Z = 25$: $[\text{Ar}]3d^5 4s^2$. Highest $n = 4 \Rightarrow$ period 4; d-block with $5 + 2 = 7 \Rightarrow$ **group 7, period 4 (manganese)**.

(c) $Z = 20$: $[\text{Ar}]4s^2 \Rightarrow$ **group 2, period 4 (calcium)**.

2.3 Nomenclature of elements with $Z > 100$

Until an official name is agreed, IUPAC builds a temporary name directly from the digits of the atomic number, using numerical roots and the suffix **-ium**.

Digit	0	1	2	3	4	5	6	7	8	9
Root	nil	un	bi	tri	quad	pent	hex	sept	oct	enn

$Z = 104 \rightarrow \text{un} + \text{nil} + \text{quad} + \text{ium} = \text{unnilquadium (Unq)}$, now rutherfordium (Rf). $Z = 120 \rightarrow \text{unbinilium (Ubn)}$.

SPELLING RULES THAT CARRY A MARK

1. "bi" and "tri" are followed by **-ium**, giving "...bium", "...trium".
2. Drop one letter where the roots would collide: "enn" + "nil" $\rightarrow$ "ennil" (not "ennnil"); "bi"/"tri" + "ium" $\rightarrow$ "bium"/"trium".
3. The symbol is the **first letter of each root**, three letters in all.

3 · Periodic Trends in Physical Properties

3.1 Effective nuclear charge — the engine behind every trend

$Z_{\text{eff}} = Z - \sigma$, where σ is the **screening (shielding) constant** due to the inner electrons. **Across a period** Z rises by one each step while the added electron enters the *same* shell and screens poorly, so Z_{eff} **increases sharply**. **Down a group** both Z and the number of screening electrons rise, so Z_{eff} stays nearly constant while the number of shells increases.

Screening efficiency falls in the order **s > p > d > f**, because a more penetrating orbital keeps the electron closer to the nucleus.

3.2 Atomic radius

An atom has no sharp boundary, so radius is always defined operationally:

- **Covalent radius** — half the internuclear distance between two identical atoms bonded by a single covalent bond (Cl_2 : 198 pm apart $\Rightarrow$ 99 pm).
- **van der Waals radius** — half the distance between the nuclei of two *non-bonded* neighbouring atoms in the solid. Always **larger** than the covalent radius, because no orbital overlap is involved.
- **Metallic radius** — half the internuclear distance between adjacent atoms in a metallic crystal.

Across a period (left $\rightarrow$ right): radius **decreases** — Z_{eff} rises while the shell number stays the same. **Down a group**: radius **increases** — a new shell is added and shielding grows.

Period 2 (pm)	Li	Be	B	C	N	O	F
Atomic radius	152	111	88	77	74	66	64

WHERE THIS LOSES MARKS

1. **Noble gases are quoted as van der Waals radii**, so they look anomalously large at the end of a period. Never use them to argue that the radius rises at the right-hand edge.
2. Across the **d-block** the decrease is very gradual and then almost flat, because the added d electron shields the outer 4s electrons rather well.

3.3 Ionic radius

Cation < parent atom — an electron (often the whole outer shell) is removed while the nuclear charge is unchanged, so the remaining electrons are pulled in harder. **Anion > parent atom** — added electrons increase repulsion with no increase in nuclear charge.

Na (186 pm) $\rightarrow$ Na⁺ (95 pm); Cl (99 pm) $\rightarrow$ Cl⁻ (181 pm).

Isoelectronic species have the same number of electrons; among them, the radius **falls as the nuclear charge rises**:

N³⁻ (171) > O²⁻ (140) > F⁻ (136) > Na⁺ (95) > Mg²⁺ (65) > Al³⁺ (50) pm — all with 10 electrons.

MEMORY HOOK

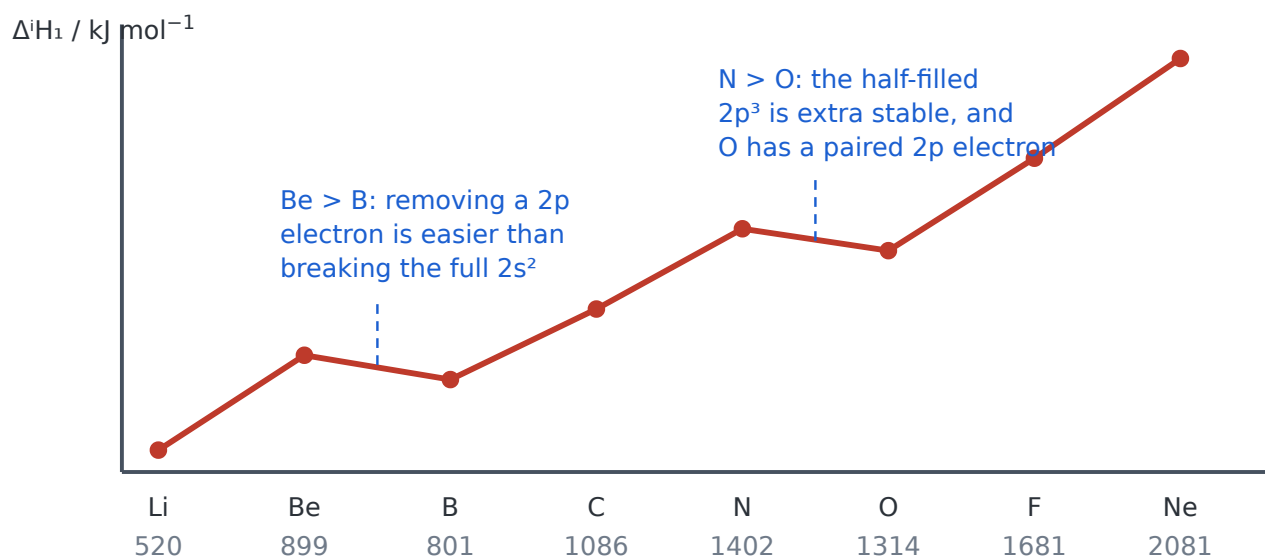
For an isoelectronic series, line the species up by **charge from most negative to most positive** — that order is already the order of decreasing size. More protons, same electrons, tighter grip.

3.4 Ionization enthalpy

$\Delta_i H_1$ is the energy required to remove the **most loosely bound electron** from an isolated **gaseous** atom in its ground state: $X(g) \rightarrow X^+(g) + e^-$. It is **always positive** (endothermic).

Factors: ionization enthalpy increases with larger Z_{eff} and decreases with larger atomic size and greater shielding; **stable half-filled and fully-filled** configurations raise it sharply.

Across a period: increases (size falls, Z_{eff} rises). **Down a group:** decreases (size rises, shielding rises). Highest in the period: the **noble gas**. Lowest: the **alkali metal**.



First ionization enthalpy across period 2. The overall rise is the Z_{eff} effect; the two dips are the whole examinable content of the graph. The identical pattern repeats in period 3 as **Mg > Al** and **P > S**.

The two anomalies, stated properly.

Be > B (and **Mg > Al**): boron's outgoing electron comes from a **2p** orbital, which is higher in energy and better shielded than beryllium's filled **2s²**, so less energy is needed.

N > O (and **P > S**): nitrogen has the **exactly half-filled 2p³** configuration, which is extra stable; oxygen's fourth 2p electron is **paired**, and the electron–electron repulsion within that orbital helps push it out.

3.5 Successive ionization enthalpies

$\Delta_i H_1 < \Delta_i H_2 < \Delta_i H_3 < \dots$ always. Removing an electron from an already positive ion is harder, because the remaining electrons are held by the same nuclear charge.

A **sudden jump** in the sequence signals that a noble-gas core has been reached, and reveals the number of valence electrons. Magnesium's third ionization enthalpy is enormous compared with its second, confirming that Mg has exactly two valence electrons and that Mg^{2+} is where it stops.

3.6 Electron gain enthalpy

$\Delta_{\text{eg}} H$ is the enthalpy change when an electron is added to an isolated gaseous atom: $\text{X(g)} + \text{e}^- \rightarrow \text{X}^-(\text{g})$. A **negative** value means energy is released — the standard convention, and the opposite in sign to "electron affinity".

$\Delta_{eg}H$ (kJ mol ⁻¹)	F	Cl	Br	I	O	S	Ne
Value	-328	-349	-325	-295	-141	-200	+116

Trends: becomes more negative across a period (halogens are the most negative of all) and less negative down a group. **Noble gases have large positive values** — the incoming electron must start a new shell.

THE CHLORINE-BEATS-FLUORINE ANOMALY — ASKED EVERY YEAR

$\Delta_{eg}H$ of **Cl is more negative than F**, and S more negative than O, even though F and O are higher in the group. **Reason:** the 2p subshell of fluorine is unusually **small and compact**, so the incoming electron suffers strong **inter-electronic repulsion** from the electrons already crowded there. Chlorine's larger 3p subshell accommodates the extra electron with far less repulsion.

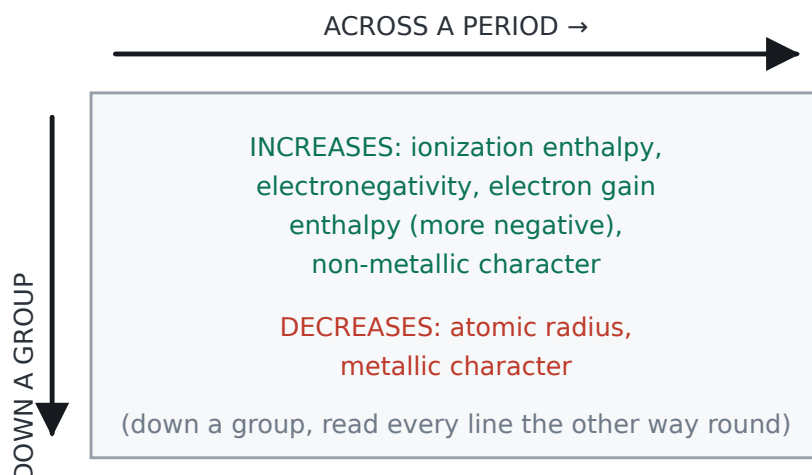
Also: the **second** electron gain enthalpy is always **positive** — forcing a second electron onto an already negative ion requires work. That is why O²⁻ formation is endothermic overall and only survives in a lattice.

3.7 Electronegativity

Electronegativity is the tendency of an atom **in a chemical compound** to attract the shared pair of electrons towards itself. It is a relative, unitless quantity — the Pauling scale sets F = 4.0, the highest of all elements.

	Electron gain enthalpy	Electronegativity
Applies to	An isolated gaseous atom	An atom within a bond
Nature	Measurable energy, in kJ mol ⁻¹	Relative number, no units

Trend: increases across a period (rising Z_{eff} , falling size) and decreases down a group. Order to remember: **F > O > N ≈ Cl > Br > C > H**. Metallic character runs exactly opposite to electronegativity.



Exceptions to memorise: Be > B, N > O, Mg > Al, P > S (ionization) · Cl > F, S > O (electron gain)

Trend compass. Everything except atomic radius and metallic character increases towards the top right of the table — which is why fluorine, at the top right of the reactive elements, is the most electronegative element there is.

4 · Periodic Trends in Chemical Properties

4.1 Valence and oxidation state

For representative elements the valence equals the number of valence electrons, or eight minus that number, whichever is smaller. Across period 3 the highest oxidation state with oxygen rises 1, 2, 3, 4, 5, 6, 7 (Na₂O, MgO, Al₂O₃, SiO₂, P₄O₁₀, SO₃, Cl₂O₇), because each of those electrons can be used in bonding.

4.2 Periodicity of oxides across period 3

Oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₃	Cl ₂ O ₇
Nature	Basic		Amphoteric	Acidic (increasingly so)			

Amphoteric means reacting with both acids and bases: Al₂O₃ + 6HCl → 2AlCl₃ + 3H₂O, and Al₂O₃ + 2NaOH → 2NaAlO₂ + H₂O. Basic character falls and acidic character rises across a period; down a group the reverse happens (so, in group 15, N₂O₅ is strongly acidic while Bi₂O₃ is basic).

4.3 Anomalous behaviour of second-period elements

Three reasons, and the exam wants all three: (i) exceptionally **small size**; (ii) high **electronegativity** and high charge/radius ratio; (iii) **absence of d orbitals** in the valence shell, so the covalency cannot exceed **four**.

Consequence: nitrogen forms NF_3 but never NF_5 , while phosphorus forms both PF_3 and PF_5 using its 3d orbitals. Oxygen's maximum covalency is 2, sulfur's is 6 (SF_6).

4.4 Diagonal relationship

The first element of a group often resembles the **second element of the next group**, diagonally down-right: **Li-Mg, Be-Al, B-Si**.

Reason: moving right across a period decreases size and increases electronegativity, while moving down a group does the opposite. Along a diagonal the two effects nearly cancel, leaving similar size, charge/radius ratio and electronegativity. Hence LiCl and MgCl_2 are both deliquescent and soluble in ethanol; Li and Mg both form nitrides directly with N_2 ; BeO and Al_2O_3 are both amphoteric.

5 · Where marks are lost

THE TEN TRAPS OF CHAPTER 3

1. **The modern periodic law is about atomic number**, Mendeleev's was about atomic mass. Quote the right one for the right scientist.
2. **Ionization enthalpy is always positive; electron gain enthalpy is usually negative**. A "large" electron gain enthalpy means large in magnitude — say "more negative" to be unambiguous.
3. **Cl has a more negative $\Delta_{eg}H$ than F**, and S than O. The reason is small size and inter-electronic repulsion in the compact 2p subshell — not electronegativity.
4. **Be > B and N > O for ionization enthalpy**. Give the configuration reason ($2s^2$ filled; $2p^3$ half-filled), never just "exception".
5. **Second electron gain enthalpy is always positive**.
6. **Isoelectronic size falls as nuclear charge rises** — N^{3-} is the biggest and Al^{3+} the smallest of the 10-electron series.
7. **Noble-gas radii are van der Waals radii**, so they are not comparable with the covalent radii of their neighbours.
8. **Zn, Cd and Hg are not typical transition metals** (d^{10} in atom and ion alike).
9. **Electronegativity has no units and is not an energy**; electron gain enthalpy is measured in kJ mol^{-1} for a gaseous atom.
10. **Hydrogen has no fixed position** — it resembles both group 1 and group 17. Say so explicitly rather than assigning it to one.

Chapter 3 of the rationalised NCERT (2023–24 onward). All ionization and electron gain enthalpies quoted in kJ mol^{-1} ; radii in pm.